Run Shi

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EDUCATION

- **Stanford University** *September* 2025 - *Present*
  M.S. in Mechanical Engineering California, United States
  - Major Focus: Thermodynamics, Mechatronics, Design and Control of Haptic Systems, Experimental Robotics, Advanced mechanical system design, Smart Product design

- **Shanghai Jiao Tong University** *September* 2021 - *June* 2025
  B.Eng. in Energy and Power Engineering Shanghai, China
  - Last two years Ranking: 3/100
  - Major Courses: Numerical Simulation of Mechanical and Power Engineering(95), Engineering Material(92), Engineering Fluid Mechanics(91), Probability and Statistics(91)

- **Karlsruhe Institute of Technology** *April* 2024 - *September* 2024
  Exchange Semester in Mechanical Engineering Karlsruhe, Germany
  - Major Courses: Heat and Mass Transfer(A+)

TECHNICAL STRENGTHS

- **Programming** Python , C++, C
- **Professional Tools & Software** GIT, CATIA, SOLIDWORKS, FEA, Creo, Fusion, UGNX, MATLAB, PCAN, MySQL, LabVIEW, OpenSAI, Ansys, Comsol, Arduino, material characterization
- **Skills** Mechanical Design & Manufacturing (DFME, FEA, DFA, prototyping design), Mechatronics (embedded system, analog circuits, digital electronics), Thermal modeling & Simulation, Material Characterization

INTERNSHIP

- **Tesla** *June* 2026 - *September* 2026
  Thermal Control Intern Palo Alto, CA
  - Developed and validated physics-based models for thermal systems, correlating simulations against bench and vehicle test data to improve thermal performance.
  - Developed estimation and control algorithms for thermal system including HVAC, heat pump to improve thermal system performance.
  - Characterize heat-exchanger and refrigeration-cycle performance, close the loop on fleet thermal control performance.

- **Tesla** *April* 2025 - *September* 2025
  High Voltage and Low Voltage Hardware Test Intern Shanghai, China
  - Supported high voltage hardware reliability test, provided data for hardware design change and implementing failure analysis.
  - Developed script implementing multi drive units reliability thermal testing, automatically uploading data to MySQL.
  - Support low voltage battery reliability test, independent test development provided data for battery management board design change.

PUBLICATIONS

- Z. Zhang, Z. Xu, **R. Shi**, C. Kim, A. Patel, C. Dai, V. R. Feig, "Water-Stable, Biodegradable, and Flexible Glycine-Based Piezoelectrics via Multi-Length-Scale Composite Engineering," Advanced Functional Materials, 2026, *e*78513. doi:10.1002/*adfm*.78513

PROJECTS

- **Target-Oriented Manipulator 7DoF Robot Arm** *May* 2026 - *June* 2026
  Stanford University Stanford, CA
  - Engineered a real-time vision-guided robotic tracking system on a 7-DoF robotic arm, combining a trained pose-detection model with an operational-space controller to autonomously detect, track, and grasp a dynamically moving target.
  - Designed a multi-state control architecture coordinating perception, motion planning, and gripper control, using filtering and smoothing techniques to ensure stable, responsive tracking despite sensor noise and a moving camera.
  - Developed a physics-based simulation environment to validate the control system and state machine end-to-end before deployment on physical hardware.

- **Autonomous Curling Robot** *February* 2026 - *March* 2026
  Stanford University Stanford, CA
  - Designed and developed a fully autonomous robot (DROID) to compete in a head-to-head curling match on a 16-foot arena, implementing precise localization via black tape navigation.
  - Engineered an innovative vane-rotation dispensing mechanism and slide system to sequentially store and accurately launch 8-oz steel pucks while adhering to a strict 13 in $\times$ 13 in starting footprint constraint.
  - Programmed a complete state machine using Arduino Mega to integrate drivetrain control and sensor fusion, ensuring seamless autonomous puck loading, scoring, and return-to-base within a 2-minute time limit.

- **Trolly for Automatic Planting of Pike Seedings in Desert** *September* 2023 - *February* 2024
  Tencent 'Light' Technology Charity Creation Camp Shanghai, China
  - Led a team in designing and prototyping an innovative automated seedling planting system for desert reforestation efforts,earning top 1% investment award.
  - Created detailed 3D models, ensuring manufacturability and on-site installation feasibility, conducted extensive field tests and iterative design improvements, adapting the system to real-world conditions.
  - Engineered and implemented a hollow drill seedling mechanism, significantly improving planting efficiency and seedling survival rates in harsh desert environments.

RESEARCH

- **Injectable In-situ Formed Piezoelectric Biocomposites with Magnetic Alignment** *Sep* 2025 - *Present*
  Stanford Univerisity Stanford, CA
  - Fabricated Janus heterostructures with single-sided magnetism by depositing ultra-thin layers of superparamagnetic nanoparticles onto microrods via Electro-hydro Dynamic (EHD) coating.
  - Designed and implemented a magnetic-assisted alignment strategy to orient microrods, overcoming the challenge of random dipole distribution and enhancing the macroscopic piezoelectric constant.
  - Developed an HSA-BB-EOH-based injectable and biocompatible composite, achieving rapid in-situ gelation in physiological environments via solvent exchange mechanisms.

- **Thermodynamic Study on Ferroelectric polymer films** *October* 2024 - *June* 2025
  Forward Interdisciplinary Research Center SJTU Shanghai, China
  - Improvement on material properties on the performance of refrigeration devices based on force-electrical-thermal coupling parameters.
  - Application of soft material property testing methods and characterization tools, including material electrocaloric effect testing, dielectric property testing, polarization property measurements, and experimental testing related to displacement and stress under electric field drive.

- **Cascade-unit Elastocaloric Cooling Regenerator** *June* 2024 - *September* 2024
  Insitute of Microstructure Technology KIT Karlsruhe, Germany
  - Engineered 3D models using Creo software to design an innovative elastocaloric experimental platform, enhancing the efficiency of solid-state cooling systems.
  - Performed finite element analysis using Ansys to optimize machine parts, ensuring they meet rigorous strength requirements while maximizing cooling performance.
  - Designed a variable cross-sectional area compression device was designed to be applicable to different cross-sectional area sizes of the elasto-thermal structure.